PHYSICS

1) According to Bohr model, energy of electron in n^{th} orbit is.....Z = atomic number.

(1) $E_n \propto \frac{n^2}{Z^2}$

(2) $E_n \propto \frac{Z^2}{n^2}$

(3) $E_n \propto \frac{n}{Z}$

(4) $E_n \propto \frac{Z}{n}$

2) Which of the following lines represents the wavelength of the spectrum of Balmer series ?

(1) $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$ where n = 3, 4...

(2) $\frac{1}{\lambda} = \frac{1}{R}\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$ where n = 3, 4...

(3) $\lambda = (2^2 - n^2)$ where n = 3, 4...

(4) Not one of these three

3) Mass defect in a $_{11}Na^{23}$ atom is 0.21 amu. Then the binding energy per nucleon will be :-

(1) 8.505 MeV

(2) 8.605 MeV

(3) 9.23 MeV

(4) 9.315 MeV

4) The binding energy per nucleon of Na^{23} is 7.68 MeV and Na^{24} is 7.48 MeV. The energy (in MeV) required to remove the extra neutron from Na^{24} is very nearly equal to :-

(1) 2.88

(2) 3.7

(3) 3.9

(4) 5

5) A photoelectric surface is illuminated successively by monochromatic light of wavelengths λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface is :

(1) $\frac{hc}{2\lambda}$

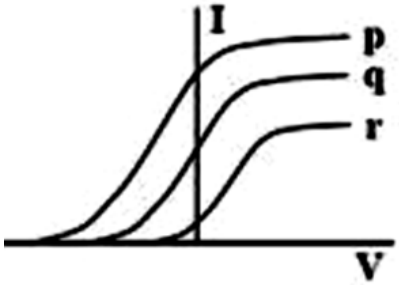
(2) $\frac{3hc}{\lambda}$

(3) $\frac{hc}{3\lambda}$

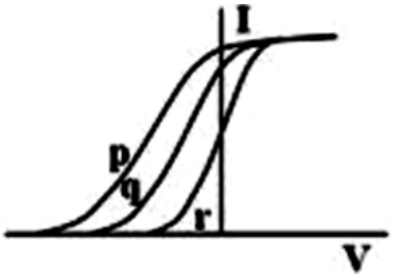
(4) $\frac{hc}{\lambda}$

6) Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions $\phi_p = 2.0eV$, $\phi_q = 2.5eV$ and $\phi_r = 3.0eV$ respectively. A light beam containing wavelengths of 550 nm, 450 nm and 250 nm with equal intensities illuminates each of the plates. The correct I - V graph for the experiment is [take hc = 1240 eV.nm]

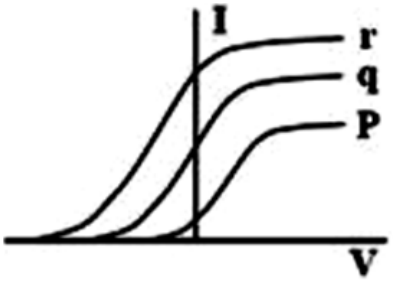
(1)



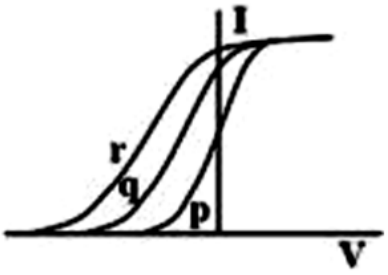
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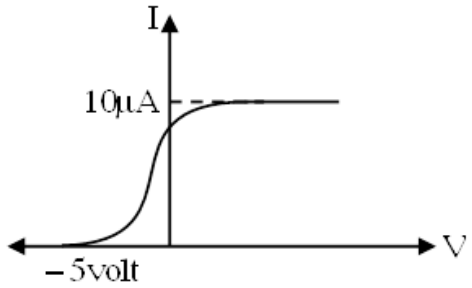


(4)



7)

In the photoelectric experiment, if we use a monochromatic light, the I–V curve is as shown. If work function of the metal is 2 eV, estimate the power of light used. (Assume efficiency of photo emission = 10^{-3} %, i.e. number of photoelectrons emitted are 10^{-3}\% of number of photons incident on metal) :-



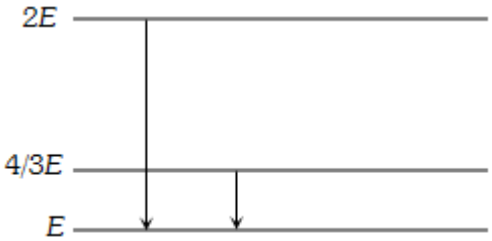
(1) 2 W

(2) 5 W

(3) 7 W

(4) 10 W

8) The following diagram indicates the energy levels of a certain atom when the system moves from $2E$ level to E, a photon of wavelength λ is emitted. The wavelength of photon produced during its transition from $\frac{4E}{3}$ level to E is



(1) $\frac{\lambda}{3}$

(2) $\frac{3\lambda}{4}$

(3) $\frac{4\lambda}{3}$

(4) 3λ

9) A stationary thoron nucleus (A = 220, Z = 90) emits an alpha particle with kinetic energy E_α. What is the kinetic energy of the recoiling nucleus ?

(1) $\frac{E_\alpha}{108}$

(2) $\frac{E_\alpha}{110}$

(3) $\frac{E_\alpha}{54}$

(4) None of these

10) Considering the Bohr model of hydrogen like atoms, the ratio of the ratio of the radius 5th orbit of the electron in Li^{2+} and He^{+} is

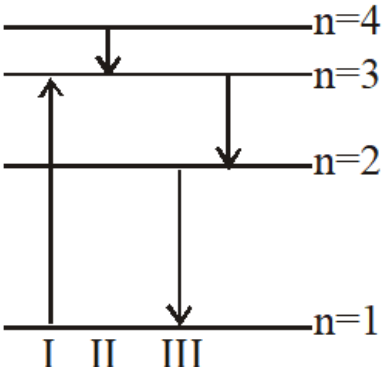
(1) $\frac{3}{2}$

(2) $\frac{4}{9}$

(3) $\frac{9}{4}$

(4) $\frac{2}{3}$

11) The diagram shows the energy levels for an electron in a certain atom. Which transition shows



represents the emission of a photon with the most energy ?

(1) IV

(2) III

(3) II

(4) I

12) The de-Broglie wavelength of a particle having kinetic energy E is λ. How much extra energy must be given to this particle so that the de-Broglie wavelength reduces to 75% of the initial value ?

(1) $\frac{1}{9}E$

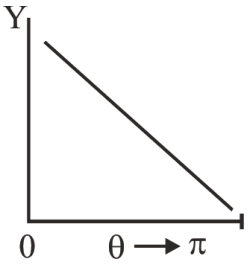
(2) $\frac{7}{9}E$

(3) E

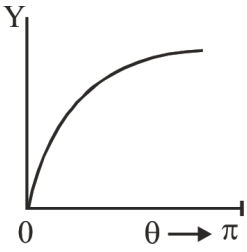
(4) $\frac{16}{9}E$

13) The graph which depicts the results of Rutherform gold foil experiment with α-particales is :
θ : Scattering angle
Y : Number of scattered a-particles detected (Plots are schematic and not to scale)

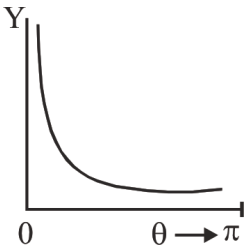
(1)



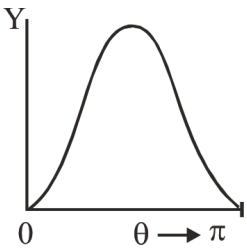
(2)



(3)



(4)



14) Light emitted during the de-excitation of electron from n = 3 to n = 2, when incident on a metal, photoelectrons are just emitted from that metal. In which of the following transition photoelectric effect is not possible for that metal?

(1) From n = 2 to n = 1

(2) From n = 3 to n = 1

(3) From n = 5 to n = 2

(4) From n = 4 to n = 3

15) A beam of light is incident normally on a plane surface absorbing 70% of the light and reflecting the rest. If the incident beam carries 10W of power, find the force exerted by its on the surface?

(1) 5.07×10^{-8}N

(2) 3.33×10^{-8}N

(3) 2.33×10^{-8}N

(4) 4.3×10^{-8}N

16) With respect to photoelectric experiment, match the entries of List-I with the entries of List-II.

List-I		**List-II**	
(P)	If ν (frequency) is increased keeping I (intensity) and ϕ (work function) constant	(1)	Stopping potential increases
(Q)	If I is increased keeping ν and ϕ constant	(2)	Saturation photocurrent increases
(R)	If the distance between anode and cathode increases.	(3)	Maximum KE of the photoelectrons increase
(S)	If ϕ is decreased keeping ν and I constant	(4)	Stopping potential remains the same

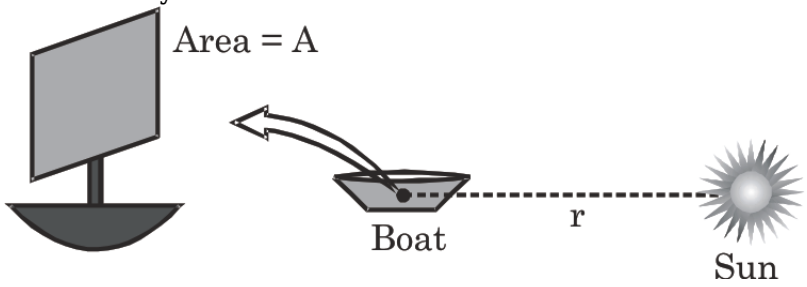
(1) P→1,3; Q→2,4; R→2; S→2,4

(2) P→1,3; Q→2,4; R→4; S→1,3

(3) P→3; Q→1,2; R→2,4; S→1

(4) P→2,3; Q→2; R→1,3; S→1,4

17) From last few decades Scientist Search alternative option to move in space. One idea is to build a boat (mass = m) with large canvas (area A) and accelerate boat by radiation force exerted by solar radiation. Find acceleration of boat when it is at distance r from sun (power of sun = P ; assume rays are normally incident on canvas and canvas is 100% reflector of light)



(1) $\frac{PA}{4\pi r^2 cm}$

(2) $\frac{PAc}{4\pi r^2 m}$

(3) $\frac{PA}{2\pi r^2 cm}$

(4) $\frac{PAc}{2\pi r^2 cm}$

18) A small plate of a metal (work function = 1.17 eV) is placed at a distance of 2 m from a monochromatic light source of wavelength 4.8×10^{-7} m and power 1.0 W. The light falls normally on the plate. Find the number of photons striking the metal plate per square metre per second:-

(1) 4.8×10^{15}

(2) 4.8×10^{16}

(3) 1.6×10^{15}

(4) 1.6×10^{16}

19) A stationary thoron nucleus (A = 220, Z = 90) emits an alpha particle with kinetic energy E_a. What is the kinetic energy of the recoiling nucleus ?

(1) $\frac{E_\alpha}{108}$

(2) $\frac{E_\alpha}{110}$

(3) $\frac{E_\alpha}{54}$

(4) None of these

20) In a certain system of unit, 1 unit of mass is 10kg, 1 unit of length is 10m and 1 unit of time is 20 sec. In this system, 1 unit of force will correspond to:

(1) 0.5 N

(2) 0.25 N

(3) 4 N

(4) None of these

21) The method of dimensional analysis can be used to derive which of the following relation:

(1) $v^2 = u^2 + 2as$

(2) $y = A \cos(\omega t - kx)$

(3) $A = A_0 e^{-\lambda t}$

(4) None of these

22) If dimensions of A and B are different, then which of the following operation is valid ?

(1) $\log\left(\frac{A}{B}\right)$

(2) $e^{A/B}$

(3) A + B

(4) A × B

23) From the point of view of significant figures. Which of the following is/are correct?
(i) 11. cm + 4 cm = 15.3 cm
(ii) 4.53m – 1.2m = 3.3 m
(iii) 5.45 kg – 3.2 kg = 2.25 kg
(iv) 84.8 cm + 48.6 cm = 133 cm

(1) ii only

(2) i, ii and iv only

(3) i, iii only

(4) ii and iv only

24) While measuring diameter of wire using screw gauge the following readings were noted. Main scale reading is 1mm and circular scale reading is equal to 42 divisions. Pitch of screw gauge is 1mm and it has 100 divisions on circular Scale. The diameter of the wire is $\frac{x}{50}$mm. The value of x is -

(1) 142

(2) 71

(3) 42

(4) 21

25) Ratio of dimension of electrical permittivity to magnetic permeability is :-

(1) $M^{-2}L^{-4}T^6A^4$

(2) $M^0L^{-2}T^2A^4$

(3) $M^0L^{-2}T^6A^4$

(4) None

26) The position of a particle at time t is given by the relation $x(t) = \left(\frac{v_0}{\alpha}\right)(1 - e^{-\alpha t})$, where v^0 is a constant and $\alpha > 0$. The dimensions of v_0 and α are respectively

(1) $M^0L^1T^{-1}$ and $M^0L^0T^{-1}$
(2) $M^0L^1T^0$ and $M^0L^0T^{-1}$
(3) $M\ L^1T^{-1}$ and $M^0L^0T^{-2}$
(4) $M^0L^1T^{-1}$ and M^0L^0T

27) In a system of units if force (F), acceleration (A) and time (T) are taken as fundamental units then the dimensional formula of energy is

(1) FA^2T
(2) FAT^2
(3) F^2AT
(4) FAT

28) A physical quantity Q is calculated according to the expression $Q = \frac{A^3B^3}{C\sqrt{D}}$. If percentage errors in A, B, C and D are 2%, 1%, 3% and 4% respectively then maximum percentage error in Q is :-

(1) 8%
(2) 10%
(3) 14%
(4) 12%

29) The period of oscillation of a simple pendulum in the experiment is recorded as 2.63 s, 2.56 s, 2.42 s, 2.71 s and 2.80 s respectively. The average absolute error is :-

(1) 0.1 s
(2) 0.11 s
(3) 0.01 s
(4) 1.0 s

30) A vernier calliper has 1 mm marks on the main scale. It has 20 equal divisions on the vernier scale which match with 16 main scale divisions. For this vernier calliper, the least count is:-

(1) 0.02 mm
(2) 0.2 mm
(3) 0.05 mm
(4) 0.1 mm

31) A science student takes 100 observations in an experiment. Second time he takes 600 observations in the same experiment. By doing so the random error becomes :-

(1) $\frac{1}{6}$ times

(2) 6 times

(3) unchanged

(4) 36 times

32) Two full turns of the circular scale of screw gauge cover a distance of 1 mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of –0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the 35th division of circular scale coincides with the reference line of main scale. The diameter of the wire is :

(1) 3.38 mm

(2) 3.32 mm

(3) 3.73 mm

(4) 3.67 mm

33) The length, breadth and thickness of a block are given by $\ell = 12\text{cm}, b = 6\text{cm}$ and $t = 2.45\,\text{cm}$. The volume of the block according to the idea of significant figures should be :-

(1) $1 \times 10^2\,\text{cm}^3$

(2) $2 \times 10^2\,\text{cm}^3$

(3) $1.763 \times 10^2\text{cm}^3$

(4) None of these

34) One centimetre on the main scale of Vernier callipers is divided into 10 equal parts if 6 divisions on main scale coincide with 10 vernier divisions then find out least count of callipers?

(1) 0.02 cm

(2) 0.01 cm

(3) 0.04 cm

(4) None of above

35) The photocurrent in experiment of photo electric effect increases if:

(1) intensity of source is increased

(2) exposure time is increased

(3) exposure time is decreased

(4) intensity of source is decreased

36) Outside a nucleus :-

(1) Neutron is stable

(2) Proton & neutron both are stable

(3) Neutron is unstable

(4) Neither neutron nor proton is stable

37) Lenard observed that no electrons are emitted when frequency of light is less than a certain minimum frequency. This minimum frequency depends on:

(1) Potential difference of emitter and collector plates

(2) Distance between collector and the emitter plate

(3) Size (area) of the emitter plate

(4) Material of the emitter plate

38) In a thermonuclear fusion, which statement is incorrect :

(1) Energy is provided by raising the temperature

(2) Energy must be provided to overcome columbic repulsion between approaching positively charged nuclei.

(3) Is responsible for the formation of red giant star

(4) Required temperature is of the order of 10^8 K.

39) How long can an electric lamp of 100W be kept glowing by fusion of 2.0 kg of deuterium? Take the fusion reaction as

$${}_1^2H + {}_1^2H \rightarrow {}_2^3He + n + 3.27\,MeV$$

(1) About 4.9×10^4 year

(2) About 9.8×10^4 year

(3) About 2.45×10^4 year

(4) About 1.64×10^4 year

40) The radius of a nucleus of mass number 64 is 4.8 fermi. Then the mass number of another nucleus having radius of 4 fermi is $\frac{1000}{x}$, where x is _____.

(1) 127

(2) 227

(3) 27

(4) 7

41)

Monochromatic light of frequency 6×10^{14} Hz
is produced by a laser. The power emitted is
2×10^{-3} W. How many photons per second on an average, are emitted by the source ?
(Given h = 6.63×10^{-34} Js)

(1) 9×10^{18}

(2) 6×10^{15}

(3) 5×10^{15}

(4) 7×10^{16}

42) In the following reaction the value of 'X' is

$_7N^{14} + _2He^4 \rightarrow X + _1H^1$

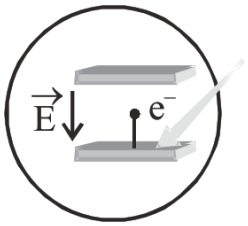
(1) $_8N^{17}$

(2) $_8O^{17}$

(3) $_7O^{16}$

(4) $_7N^{16}$

43) The collector plate in an experiment on photoelectric effect is kept vertically above the emitter plate. Light source is put on and a saturation photo current is recorded. An electric field is switched on which has a vertically downward direction. Then :



(1) The photo current will increase

(2) The kinetic energy of the electrons will increase

(3) The stopping potential will decrease

(4) The threshold wavelength will increase

44) Assertion : The energy (E) and momentum (p) of a photon are related by p = E/c.
Reason : The photon behaves like a particle.

(1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(3) If the Assertion is correct but Reason is incorrect.

(4) If both the Assertion and Reason are incorrect.

45) In the Bohr model of a hydrogen atom, the centripetal force is furnished by the coulomb attraction between the proton and the electron. If a_0 is the radius of the ground state orbit, m is the mass, e is the charge on the electron and ε_0 is the vacuum permittivity, the speed of the electron is

(1) 0

(2) $\frac{e}{\sqrt{\varepsilon_0 a_0 m}}$

(3) $\frac{e}{\sqrt{4\pi\varepsilon_0 a_0 m}}$

(4) $\frac{\sqrt{4\pi\varepsilon_0 a_0 m}}{e}$

CHEMISTRY

1) Isoelectronic species, in which first one is larger in size than the second :

(1) P^{-3}, N^{-3}

(2) Cl^{-}, S^{-2}

(3) Ca^{+2}, Ti^{+4}

(4) S^{-2}, P^{-3}

2) In which of the following, Mn will have small size ?

(1) MnO

(2) MnO_2

(3) MnO_3

(4) Mn_2O_3

3) First four I.P. values of an element are 19, 190, 230 and 260 ev/atom. Then element is :

(1) Li

(2) Al

(3) F

(4) Na

4) Amongest the elements with following electronic configurations, which one of them may have the highest ionization energy ?

(1) $Ne[3s^2 3p^1]$

(2) $Ne[3s^2 3p^3]$

(3) $Ne[3s^2 3p^2]$

(4) $Ar[3d^{10} 4s^2 4p^3]$

5) IP of Be is higher than :
(a) B (b) Mg (c) Li
(d) Al (e) K

(1) a and b

(2) a, b and d

(3) a, b, d and e

(4) a, b, c, d and e

6) Which of the following electron affinity order is not correct ?

(1) Be < N < B < O

(2) Al < P < Si < S

(3) N < P < C < Si

(4) P < N < Si < C

7) An ion X^{-} has configuration $2s^2 2p^6$, which is same for other ion Y^{+}. Then :

(1) IP of X = IP of Y

(2) EA of X = EA of Y

(3) IP of X > EA of Y^+

(4) IP of X^- > IP of Y^+

8) Which is a metalloid ?

(1) Pb

(2) Sb

(3) Bi

(4) Zn

9) Amphoteric oxide is ?

(1) CO_2

(2) Al_2O_3

(3) CaO

(4) SiO_2

10) The ionization energy order is :

(1) K < Na < Si < Cl < S

(2) K < Na < Si < S < Cl

(3) Si < Na < Cl < K < S

(4) Na < K < Si < S < Cl

11) For Na^+, Mg^{2+}, F^- and O^{2-}; the correct order of increasing ionic radii is

(1) $Mg^{2+} < Na^+ < F^- < O^{2-}$

(2) $O^{2-} < F^- < Na^+ < Mg^{2+}$

(3) $Na^+ < Mg^{2+} < F^- < O^{2-}$

(4) $Mg^{2+} < O^{2-} < Na^+ < F^-$

12) The elements with atomic numbers 35, 53 and 85 are all _____.

(1) noble gases

(2) halogens

(3) heavy metals

(4) light metals

13) Which of the following is different from other three oxides :-

(1) CO

(2) NO

(3) Na_2O

(4) N_2O

14) ***Assertion :-*** Boron has a smaller first ionisation enthalpy than beryllium.
Reason :- The penetration of a 2s electron to the nucleus is more than the 2p electron

(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

15) Configuration $[Rn]6d^2 7s^2$ belong to :-

(1) 7th period, 2nd group, s-block
(2) 7th period, 4th group, d-block
(3) 7th period, 14th group, p-block
(4) 7th period, 3rd group, f-block

16) Correct order of metallic character is :-

(1) P < Si < Be < Mg < Na
(2) Si < P < Be < Na < Mg
(3) P < Si < Mg < Be < Na
(4) Si < P < Mg < Na < Be

17) The IP_1, IP_2, IP_3, IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be:-

(1) Na
(2) Si
(3) F
(4) Ca

18) IP_1 and IP_2 of Mg are 178 and 348 K. cal mol^{-1}. The enthalpy required for the reaction $Mg \rightarrow Mg^{2+} + 2e^-$ is :-

(1) + 170 K.cal
(2) + 526 K.cal
(3) – 170 K.cal
(4) – 526 K.cal

19) The screening effect of d- electrons is :-

(1) Equal to the p - electrons
(2) Much more than p - electrons
(3) Same as f - electrons
(4) Less than p – electrons

20) Which of the following is most basic oxide?

(1) BeO

(2) MgO

(3) CaO

(4) SrO

21) Which of the following will have the most negative electron gain enthalpy and which the least negative?

(1) F, CI

(2) CI, F

(3) CI, S

(4) CI, P

22) Which of the following is not affected by stable configuration ?

(1) Ionisation potential

(2) Electron affinity

(3) Electronegativity

(4) All

23) Among the following the correct order of acidic strength is

(1) $Al_2O_3 < SiO_2 < SO_3 < P_2O_5$

(2) $SiO_2 < SO_3 < Al_2O_3 < P_2O_5$

(3) $SO_3 < P_2O_5 < SiO_2 < Al_2O_3$

(4) $Al_2O_3 < SiO_2 < P_2O_5 < SO_3$

24)

	Column A (Atomic no./configuration)		**Column B (Block)**
a.	105	i.	p
b.	117	ii.	d
c.	100	iii.	f
d.	[Rn] $7s^2 5f^{14} 6d^{10}$	iv.	s
e.	[Rn] $7s^2 5f^{14} 6d^1$		

Select the correct match :-

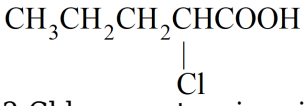
(1) a→(i), b→(i), c→(ii), d→(ii), e→(iii)

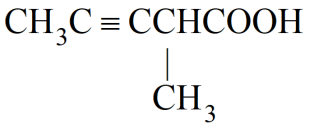
(2) a→(ii), b→(i), c→(iii), d→(ii), e→(iii)

(3) a→(ii), b→(i), c→(iii), d→(iii), e→(ii)

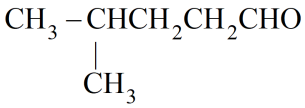
(4) a→(ii), b→(ii), c→(iii), d→(iii), e→(ii)

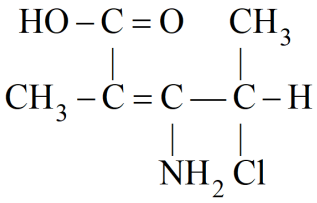
25) Which of the following compound is wrongly named?

(1) 
2-Chloro pentanoic acid

(2) 
2-Methyl hex-3-enoic acid

(3) $CH_3CH_2CH{=}CHCOCH_3$
Hex-3-en-2-one

(4) 
4-Methyl pentanal

26) The IUPAC name of compound 

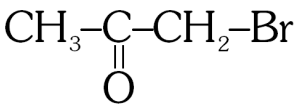
(1) 2-Amino-3-chloro-2-methylpent-2-enoic acid

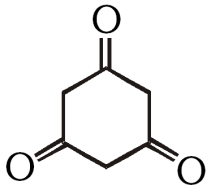
(2) 3-Amino-4-chloro-2-methylpent-2-enoic acid

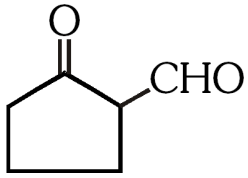
(3) 4-Amino-3-chloro-2-methylpent-2-enoic acid

(4) All of the above

27) Which is correct IUPAC name :-

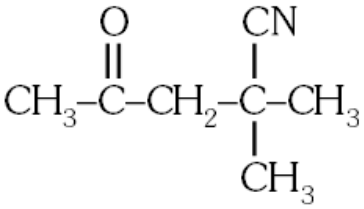
(1) 
1 -Bromopropanone

(2) 
1,3,5-Cyclohexanetrione

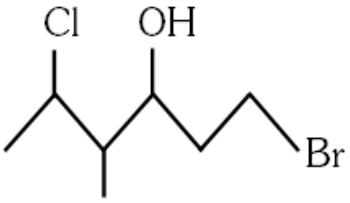
(3) 
2-Ketocylopentanecarbaldehyde

(4) All are correct

28) The IUPAC name of the compound

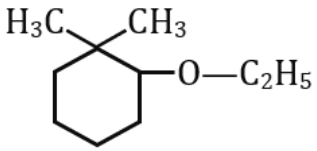


(1) 2-Cyano-2-methyl-4-oxopentane
(2) 4-Cyano-4-methyl-2-pentanone
(3) 4-Cyano-4-methyl-2-oxopentane
(4) 2, 2-Dimethyl-4-oxopentanenitrile



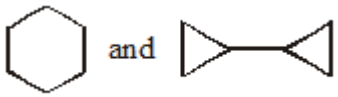
29) The correct IUPAC name of the following compound is :

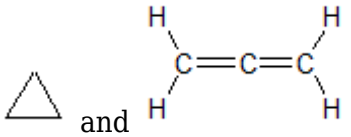
(1) 6-bromo-2-chloro-4-methylhexan-4-ol
(2) 1-bromo-4-methyl-5-chlorohexan-3-ol
(3) 6-bromo-4-methyl-2-chlorohexan-4-ol
(4) 1-bromo-5-chloro-4-methylhexan-3-ol

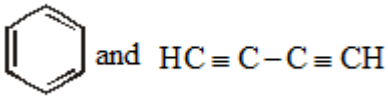
30)  IUPAC name of given compound is

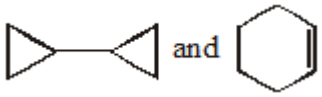
(1) 1-Methoxy-2, 2-dimethyl cyclohexane
(2) 2-Ethoxy-1, 1-dimethyl cyclohexane
(3) 1-Ethoxy-2, 2-dimethyl cyclohexane
(4) 2-Methoxy-1, 1-dimethyl cyclohexane

31) The pair of compounds having the same molecular formula

(1) 

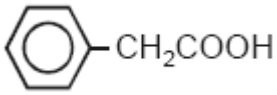
(2) 

(3) 

(4)  and

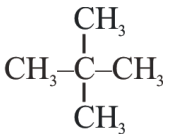
32) IUPAC name of $(CH_3)_2CHCH(CH_3)_2$ is :-

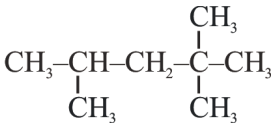
(1) 2,2-Dimethylbutane
(2) 2,3-Dimethylbutane
(3) 2,4-Dimethylbutane
(4) 1-Methylpentane

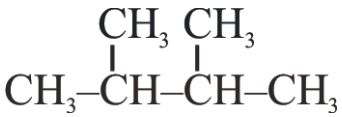
33) IUPAC name for  is

(1) phenylacetic acid
(2) 2-phenylethanoic acid
(3) benzylformic acid
(4) benzoacetic acid

34) The prefix 'Neo' is not used for :–

(1) 

(2) 

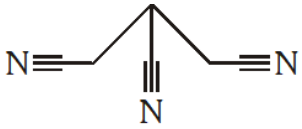
(3) 

(4) 2 & 3 Both

35) Number of π bonds in ethylene chloride.

(1) 0
(2) 1
(3) 2
(4) 3

36) IUPAC name of



(1) Propane-1, 2, 3-tricarbonitrile
(2) 3-Cyanopentane-1, 5-dinitrile

(3) Propane-1, 2, 3-tricyanide

(4) 2-Cyanopropane-1, 3-dinitrile

37) IUPAC name of  is-

(1) 3-Ethyl-6, 6-dimethyl cyclohexene

(2) 4-Ethyl-1, 1-dimethyl cyclohex-2-ene

(3) 6-Ethyl-3, 3-dimethyl cyclohexene

(4) 1-Ethyl-4, 4-dimethyl cyclohex-2-ene

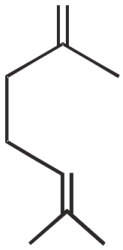
38) Unsaturated hydrocarbon is :

(1) $CH_3–CH_2–CH_2–CH_3$

(2) $CH_3–CH=CH–CH_3$

(3) $CH_3–CH_2–COOH$

(4) $CH_3–CH_2–CHO$

39) IUPAC name of the compound  is :-

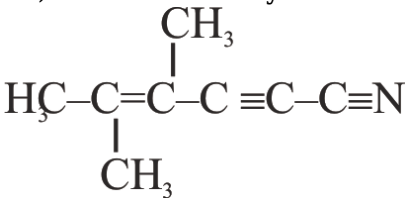
(1) 1,1,5-Trimethyl-1, 6-hexadiene

(2) 2,6-Dimethyl-2, 6-heptadiene

(3) 2,6-Dimethylhepta-1,5-diene

(4) None of these

40) Number of acetylenic bond in the given compound is :-

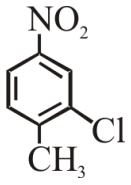


(1) 1

(2) 2

(3) 3

(4) 4

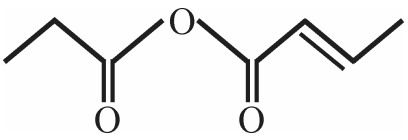
41) Correct IUPAC name of  is :-

(1) 4–methyl–5–chloro nitrobenzene

(2) 2–chloro–1–methyl–4–nitrobenzene

(3) 4–methyl–3–chloro–1–nitrobenzene

(4) 6–chloro–1–methyl–4–nitrobenzene

42) IUPAC name of compound 

(1) But-1-en-oic propanoic anhydride

(2) Propanoic but-2-en-oic anhydride

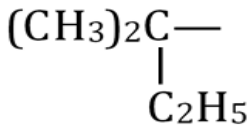
(3) Propanoic butanoic-2-en-anhydride

(4) But-2-en-oic propanoic anhydride

43) Which one is a tertiary radical (3° radical) :-

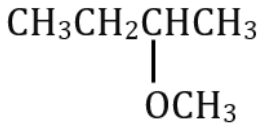
(1) $(CH_3CH_2)_2CH—$

(2) $(CH_3)_2C—CH_2—$

(3) 

(4) $(CH_3)_2CH—$

44) A tertiary ether of molecular formula $C_5H_{12}O$ is

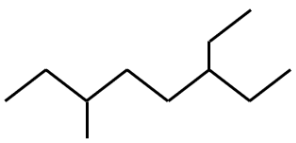
(1) 

(2) $(CH_3)_3COCH_3$

(3) $CH_3CH_2OCH(CH_3)_2$

(4) $CH_3CH_2OCH_2CH_2CH_3$

45) IUPAC name of the given compound is :



(1) 3-methyl-6-ethyloctane

(2) 3-ethyl-6-methyloctane

(3) 3,6-dimethyloctane

(4) 2,5-diethylheptane

BIOLOGY

1) Study the following statements carefully :-
(A) All plants, animals, fungi and microbes exhibit metabolism
(B) Perhaps, the most obvious and technically complicated feature of all living organisms is the ability to sense their surroundings or environment and respond to these stimuli
(C) All organisms have consciousness but the human has self consciousness and not the consciousness
(D) Living organisms are self replicating, and evolving but not self regulating
Find the most correct option from the followings:

(1) A, B and C are correct and D is incorrect

(2) A and B are correct and C and D are incorrect

(3) A and B are incorrect and C and D are correct

(4) A is correct and B, C and D are incorrect

2) _____ can not be an all inclusive defining characteris of living organisms :-

(1) Sexual reproduction

(2) Cellularity

(3) Metabolism

(4) Responding nature

3) Read the statements and identify A, B, C & D :-
(1) Cell is the ___(A)___ unit of life.
(2) Cell is ___(B)___ and ___(C)___ unit of all living organism.
(3) Ability to sense the surrounding environment and respond to environmental stimuli is called ___(D)___.

	A	B	C	D
(1)	Fundamental structural	Consciousness	Basic	Functional
(2)	Basic	Consciousness	Fundamental structural	Functional
(3)	Consciousness	Basic	Functional	Fundamental structural
(4)	Basic	Fundamental structural	Functional	Consciousness

(1) 1

(2) 2

(3) 3

(4) 4

4) In Potato, Brinjal, Makoi, Lion & leopard. How many species, genera and families are respectively ?

(1) 5, 3, 2
(2) 4, 3, 2
(3) 5, 2, 2
(4) 4, 3, 2

5) *Mangifera*, *Triticum* and *Homo* are:-

(1) Same taxa at same level
(2) Different taxa at different level
(3) Different taxa at same level
(4) Same taxa at different level

6)

Match the following columns.

	Column-I		**Column-II**
A.	Family	1.	*Mangifera indica*
B.	Kingdom	2.	Polymoniales
C.	Order	3.	*Solanum*
D.	Species	4.	Plantae
E.	Genus	5.	Solanaceae

	A	**B**	**C**	**D**	**E**
(1)	5	4	2	1	3
(2)	5	4	2	3	1
(3)	1	2	3	5	4
(4)	2	1	3	4	5

(1) 1
(2) 2
(3) 3
(4) 4

7) The system of naming animals based on agreed principles and criteria is known as:

(1) International Code for Botanical Nomenclature (ICBN)
(2) International Code for Zoological Nomenclature (ICZN)

(3) International Code for Microbial Nomenclature

(4) International Code for Genetic Nomenclature

8) Arrange the following categories in descending order for the classification of mango.
(i) Dicotyledonae
(ii) Anacardaceae
(iii) Angiospermae
(iv) Sapindales

(1) (i) - (ii) - (iv) - (iii)

(2) (iii) - (i) - (ii) - (iv)

(3) (iii) - (i) - (iv) - (ii)

(4) (ii) - (iv) - (i) - (iii)

9) Select the wrong statements.
(i) Lower the taxon, more are the characteristics that the members within the taxon share.
(ii) Order is assemblage of genera which exhibit a few similar characters.
(iii) Cat and Dog are included in the same family Felidae.
(iv) Binomial nomenclature was introduced by Carolus Linnaeus.

(1) i, ii and iii

(2) ii, iii and iv

(3) i and iv

(4) ii and iii

10) **Assertion :** Complexity of classification increases from kingdom to species.
Reason : Common characters decreases from kingdom to species.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are correct, but Reason is the not correct explanation of Assertion.

(3) Assertion is correct, but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

11) Ascending or descending arrangement of taxonomic categories is known as :-

(1) Identification

(2) Key

(3) Taxonomy

(4) Hierarchy

12) Taxonomy based on cytological information on cell, chromosome number, their structure and behaviour during meiosis is–

(1) Adansonian taxonomy

(2) Cytotaxonomy

(3) Chemotaxonomy

(4) Ω-taxonomy

13) The most convenient method of nomenclature system, used for nomenclature of maximum plants is :-

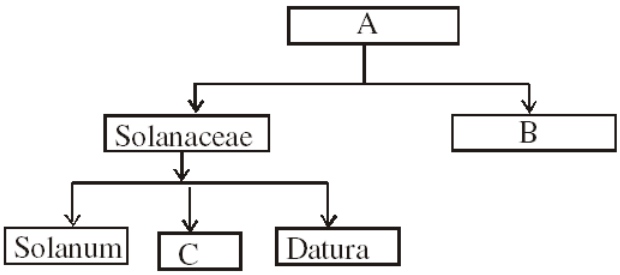
(1) Polynomial nomenclature
(2) Binomial nomenclature
(3) Trinomial nomenclature
(4) Both (1) & (2)

14)

Which is correct in nomenclature :

(1) Both genus and specific epithet are printed in italics
(2) Genus & specific epithet may be same words in ICBN
(3) Both in genus and specific epithet, the first letter is capital
(4) Genus is written after the name of specific epithet

15) Recognise the following diagram & find the correct option:



Here A, B & C are : -

(1) A- Polymoniales , B- Fabaceae, C- Brassica
(2) A - Convulvulaceae , B- Petunia , C- Polymoniales
(3) A - Fabaceae , B- Petunia , C- Convulvulaceae
(4) A- Polymoniales , B- Convulvulaceae , C- Petunia

16) A collection of species, which bears a close resemblance to one another in the morphological characters of the floral parts is known as :-

(1) Family
(2) Genus
(3) Division
(4) Variety

17) Consider the following taxa and select how many of them do not belong the lowest cotegory of taxonomic hierarchy ?
Chili, Lion, Dicots, Animals, Mango, Sweet potato, Polymoniales, Cat

(1) Seven
(2) Six
(3) Five

(4) Three

18) Which of the following kingdoms were added after the 2-kingdom system of Linnaeus
(A) Monera
(B) Protista
(C) Fungi
(D) Plantae
(E) Animalia

(1) A, B and C
(2) B, C and D
(3) C, D and E
(4) All

19) Correct scientific name of Mango is–

(1) *Mangifera Indica*
(2) *Mangifera indica* Linn.
(3) *Mangifera mangifera*
(4) *Solanum mangifera*

20) Nomenclature is governed by certain universal rules. Which one of the following is contrary to the rules of nomenclature?

(1) Biological names can be written in any language.
(2) The first word in a biological name represents the genus name, and the second is a specific epithet.
(3) The names are written in Latin and are italicised.
(4) When written by hand, the names are to be underlined separately.

21)

Maximum no. of common characters between spirogyra and mango will be at the level of

(1) Class
(2) Family
(3) Kingdom
(4) Order

22) In a hierarchical system of plant classification, which one of the following taxonomic ranks generally ends in ‘aceae’

(1) Family
(2) Genus
(3) Order
(4) Class

23) Taxonomically a species is :-

(1) A group of individual organisms with fundamental similarities
(2) A group of individuals capable of interbreeding
(3) A group of individuals living together in a habitat
(4) A group of individuals having same morphology only

24) Mayr's biological concepts of species is mainly based on :-

(1) Morphological traits
(2) Reproductive isolation
(3) Modes of reproduction
(4) Morphology and reproduction

25) In Whittaker's five kingdom system of classification, eukaryotes are distributed among :-

(1) Four kingdoms
(2) Three kingdoms
(3) Two kingdoms
(4) Five kingdoms

26) **Assertion :** Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.
Reason : Characterisation, identification, classification and nomenclature are the processes that are basic to taxonomy.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

27) Odd one w.r.t. same taxa

(1) Poales, Sapindales, Polymoniales
(2) Cat, Dog, Wheat
(3) Chordata, Thallophyta, Gymnosperm
(4) Animal, Mammals, Panthera

28) **Statement-I :** Each category, referred to as a unit of classification infact, represents a rank and is commonly termed as taxa.
Statement-II : These taxonomic groups/categories are distinct biological entities and not merely morphological aggregates.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

29) In the five kingdom classification of whittaker there is no mention of :-

(1) Lichen, Virus, Viroids and bacteria
(2) Virus, viroids, Prion and algae
(3) Prions, viroids, viruses and lichen
(4) Algae, Bacteria, Lichen and Virus

30)

Which statement(s) is/are not true for five kingdom classification proposed by R.H Whittakar [1969]-
(a) Main criteria for classification used by him includes cell structure, thallus organization, mode of nutrition. Reproduction and phylogenetic relationship
(b) Chlamydomanas, chlorella and spirogyra are placed in algae within plants due to presence of cell wall
(c) Fungi are placed in a separate kingdom-kingdom fungi
(d) Paramecium and amoeba are placed in Animal kingdom due to due to lack of cell wall
(e) All prokaryotic organisms such as bacteria, BGA and ferns are grouped together under kingdom monera.

(1) Only a
(2) a, b and d
(3) a, d and e
(4) b, d and e

31) In Carl Woese's classification system, the kingdom that includes fungus-

(1) Bacteria
(2) Eukarya
(3) Monera
(4) Archaea

32) Which of the following kingdom exhibit heterotrophic like holozoic and saprophytic mode of nutrition only :-

(1) Monera
(2) Plantae
(3) Fungi
(4) Animalia

33) According to three domain, how many kingdom include prokaryotes only ?

(1) Three
(2) Two
(3) One
(4) None

34) Five kingdom system of classification suggested by R.H. Whittaker is not based on :-

(1) Complexity of body organisation
(2) Presence or absence of a well-defined nucleus
(3) Mode of locomotion
(4) Mode of nutrition

35) First of all the plants are classified into herbs, Shrubs and tree by :

(1) Linnaeus
(2) Aristotle
(3) R.H. Whittakar
(4) Copeland

36) Earlier classification systems included bacteria, blue green algae, fungi, mosses, ferns, gymnosperm and the angiosperms under plants. Which character are taken for this classification ?

(1) Cell membrane
(2) Cell wall
(3) Mode of nutrition
(4) Type of reproduction

37) Identify the correct option for A, B, C regarding taxonomic categories of wheat.
Triticum → A → Poales → B → Angiosperm → C.

(1) A-Aestivum, B-Dicotyledonae, C-Plantae
(2) A-Poaceae, B-Monocotyledonae, C-Plantae
(3) A-Monocotyledonae, B-Aestivum, C-Dicotyledonae
(4) A-Polymoniales, B-Monocotyledonae, C-Plantae

38) Read the following statements and select the correct ones.
(i) Increase in mass and increase in number of individuals are twin characteristics of growth
(ii) In unicellular organisms growth and reproduction are same.
(iii) 'Response to stimuli' is a defining property of living organisms.

(1) (i) and (ii)
(2) (ii) and (iii)
(3) (i) and (iii)
(4) (i), (ii) and (iii)

39) Which of the following is not correct for classification

(1) Carolus Linnaeus classification is an artificial classification
(2) Large number of characters can be used in numerical classification
(3) Equal weightage is given to vegetative and reproductive characters in natural classification
(4) All of the above

40) According to five kingdom classification photosynthetic organism belongs to how many kingdoms ?

(1) Only one kingdom
(2) Two kingdoms
(3) All five kingdoms
(4) Three kingdoms

41) **Statement I :** Genus comprises a group of related species which has less characters in common in comparison to species of other genera.
Statement II : Family, has a group of related genera with still more number of similarities as compared to genus and species.

(1) Both statements are correct
(2) Statement I is correct and II is incorrect
(3) Statement I is incorrect and II is correct
(4) Both statements are incorrect

42) R. H. Whittekar 5 kingdom classification bought :

(1) Bacteria with fungi
(2) Chlamydomonas & spirogyra with algae
(3) Paramecium & amoeba with chlorella
(4) Spirogyra & Paramecium with chlorella

43) What is/are demerits of two kingdom system of classification?

(1) It did not distinguish prokaryotes and eukaryotes
(2) It did not distinguish unicellular and multicellular organisms
(3) It did not separate photosynthetic and non-photosynthetic organisms
(4) All of the above

44) Who among the following was the earliest to attempt a more scientific basis of classification-

(1) John Ray
(2) R.H. Whittaker
(3) Aristotle
(4) Linaeus

45) Select the correct statement
A. In three domain system, prokaryotes are present in two domain
B. In three domain system, eukaryotes are present in single domain
C. Phylogeny was not the criteria used by R.H. Whittaker.
D. Body organisation is cellular in kingdom Monera.

(1) Only A and B
(2) B, C and D

(3) A, B and D

(4) A, B, C and D

46) (A) _____ Echinoderms are marine
(B) _____ Porifera are symmetrical
(C) _____ Ctenophora are marine
(D) _____ Annelids are parasitic
How many above blanks can be fill will “All” word

(1) 1

(2) 2

(3) 3

(4) 5

47) Water vascular system is the characteristic of which group of the following:

(1) Porifera

(2) Ctenophora

(3) Echinodermata

(4) Chordata

48) Segmentation in the body is first observed in which of the following:

(1) Platyhelminthes

(2) Aschelminthes

(3) Annelida

(4) Arthropoda

49) Which of the following statement is **incorrect** about metagenesis-

(1) Alternation of asexual and sexual phases in the life cycle of obelia is called metagenesis.

(2) Polyp produces medusa asexually and medusa produces polyp by sexual means.

(3) Medusa is the sexual phase and polyp is the asexual phase.

(4) Both polyp and medusas are sessile

50) Which of the following set of animals are hermaphrodite-

(1) Taenia, Pheretima, fasciola

(2) Fasciola, Taenia, Ascaris

(3) Fasciola, Ascaris and Pheretima

(4) Pleurobrachia, Fasciola and Ascaris

51) Worm-like marine animals and their body divided into proboscis, collar and trunk as well as open type circulation seen in which of the following animal-

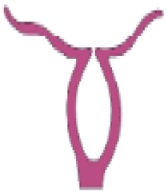
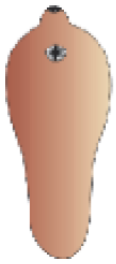
(1) Saccoglossus

(2) Ascaris

(3) Wuchereria

(4) Ascidia

52) Choose the correct match from column-I, II and III :

	Column-I		Column-II		
(A)		(i)	Platyhelminthes	(a)	Tissue level of organisation
(B)		(ii)	Aschelminthes	(b)	Organ system level of organisation
(C)		(iii)	Porifera	(c)	organ level of organisation
(D)		(iv)	Coelenterata	(d)	Cellular level of organisation

(1) A-i-a, B-ii-b, C-iii-c, D-iv-d

(2) A-ii-b, B-iv-a, C-i-c, D-iii-d

(3) A-iii-c, B-i-d, C-ii-a, D-iv-b

(4) A-iv-d, B-iii-c, C-ii-b, D-i-a

53) Match column-I with column-II and select correct option about the :-

Column-I		**Column-II**	
(A)	Spongilla	(i)	Sea fan
(B)	Gorgonia	(ii)	Brain coral
(C)	Euspongia	(iii)	Fresh water sponge
(D)	Meandrina	(iv)	Sea Pen
(E)	Pennatula	(v)	Bath sponge

(1) A-i, B-ii, C-iii, D-iv, E-v

(2) A-iii, B-i, C-v, D-ii, E-iv

(3) A-ii, B-iv, C-i, D-iii, E-v

(4) A-iii, B-v, C-iv, D-i, E-ii

54) What is the right sequence of structure of mollusca from outside to inside -

(1) Shell → Mantle cavity → Mantle → Visceral hump.
(2) Mantle → Shell → Mantle cavity → Visceral hump.
(3) Shell → Mantle → Mantle cavity → Visceral hump.
(4) Shell → Mantle → Visceral hump. → Mantle cavity

55) Which of the following animal absorbs nutrients from the host directly through their body surface?

(1) *Sycon*
(2) *Planaria*
(3) *Taenia*
(4) *Limulus*

56) Choose the incorrect set w.r.t animal group, its habitat and characteristic feature.

Animal group, Habitat, Feature

(1) Echinoderms, Marine, Water vascular system
(2) Ctenophores, Marine, Ciliated comb plates
(3) Flatworms, Parasitic, Flame cells for excretion
(4) Molluscs, Terrestrial, Stomach contains radula

57) Consider the following features.
(a) Organ-system level of organisation
(b) True coelomates
(c) Metamerically segmented
Select the correct phylum which possess all the above characteristics.

(1) Mollusca
(2) Annelida
(3) Hemichordata
(4) Echinodermata

58) Read the following statements and select the correct option.
Statement-A: The property of a living organism to emit light is well marked in comb jellies.
Statement-B: *Taenia* exhibits true metameric segmentation.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only A is correct
(4) Only statement B is correct

59) Choose the mismatch pair w.r.t the organism and its excretory structure.

(1) *Ophiura-Gills*

(2) *Anopheles*-Malphigian tubules

(3) *Pheretima*-Nephridia

(4) *Planaria-Flame cells*

60) Identify the number of organisms from the box given below which are hermaphrodites and have internal fertilisation.

Pleurobrachia, Nereis, Laccifer, Spongilla, Fasciola

(1) Five

(2) Two

(3) Four

(4) Three

61) Which of the following is a gregareous pest ?

(1) *Pheretima*

(2) *Culex*

(3) *Locusta*

(4) *Ascaris*

62) Which one of the following is not a characteristic of phylum-Annelida?

(1) Closed circulatory system

(2) Segmentation

(3) Pseudocoelom

(4) Ventral nerve cord

63) Match the following :

Column-I		**Column-II**	
(a)	Physalia	(i)	Pearl oyster
(b)	Limulus	(ii)	Portuguese Man of War
(c)	Ancylostoma	(iii)	Living fossil
(d)	Pinctada	(iv)	Hookworm

Choose the correct answer from the options given below :

	(a)	(b)	(c)	(d)
(1)	ii	iii	iv	i
(2)	i	iv	iii	ii
(3)	ii	iii	i	iv
(4)	iv	i	iii	ii

(1) 1

(2) 2

(3) 3

(4) 4

64) Choose the correct option

	Scientific name	Common name	Features
(1)	Hirudinaria	Blood sucking leech	- Unsegmented body - Closed circulation
(2)	Laccifer	Lac insect	- Calcareous exoskeleton - Open circulation
(3)	Dentalium	Tusk shell	- Unsegmented body - Calcareous exoskeleton
(4)	Antedon	Sea urchin	- Water vascular system - Spiny skined

(1) 1

(2) 2

(3) 3

(4) 4

65) **Assertion :** Excretory system is absent in Echinodermata.
Reason : Echinodermata are exclusively marine.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

66) **Assertion :** Radula found in Phylum-Mollusca and helps in feeding.
Reason : Radula is file-like rasping organ.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

67) Mark the option which represents animals from the largest and the second largest phylum of animal kingdom respectively.

(1) Aplysia, Laccifer

(2) Bombyx, Loligo

(3) Sepia, Aedes

(4) Culex, Ophiura

68) Which of the following is incorrect match?

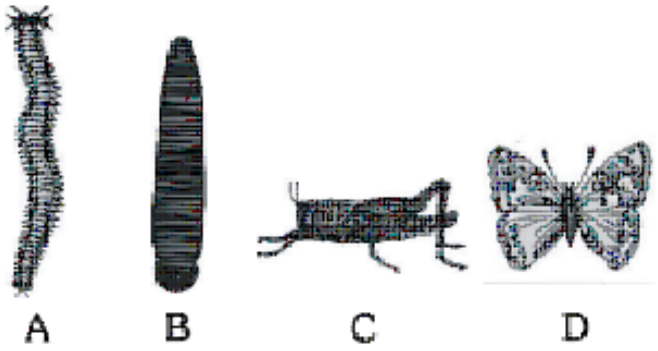
(1) Vector - Aedes

(2) Gregarious pest - Locusta

(3) Living fossil - Limulus

(4) Economically beneficial insect - Aphid

69) Which of the following animals illustrated in diagram given below is/are dioecious, coelomate, with metameric segmentation and closed circulatory system?



(1) A only

(2) A and B

(3) A and C

(4) B and D

70) Read the following statements and give the correct answer:

Phylum	Character	Examples
(1) Annelid	Nephridia for osmoregulation & excretion	Nereis, Ancylostoma
(2) Mollusca	Body segmented	Pinctada, Aplysia
(3) Echinodermata	sexual reproduction	Cucumaria Antedon
(4) Cnidaria	Presence of Paragastric cavity	Meandrina, Limulus

(1) 1

(2) 2

(3) 3

(4) 4

71) **Assertion :** Tissues are grouped together to form organs, each specialised for a particular function.
Reason : Organ level is exhibited by members of Platyhelminthes.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

72) **Statement-I :-** The body cavity, which is lined by mesoderm is called coelom.
Statement-II :- Animals possessing coelom are called coelomates.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

73)

Type of fertilization in sponges is :

(1) Self, internal
(2) Cross, internal
(3) Cross, external
(4) Self, external

74) Diploblastic and triploblastic are terms that describe

(1) The number of invaginations during embryonic development
(2) The number of heads during embryonic development
(3) The number of germinal layers during embryonic development
(4) The number of cell types during development

75) **Statement - I :** In open circulation blood is pumped out of heart and the cell and tissues are directly bathed in it.
Statement - II : In close circulation blood is circulated through series of vessles of varying diameters.

(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is incorrect and statement-II is correct.
(4) Statement-I is correct and statement-II is incorrect.

76) Which of the following is correct for medusa ?

(1) Sessile
(2) Cylindrical
(3) Free swimming
(4) Present in Hydra

77) Tissue level of body organisation are present in ____

(1) *Euspongia*

(2) *Spongilla*

(3) *Taenia*

(4) *Obelia*

78) Soft body and Presence of Mantle are the characteristic feature of :-

(1) Arthropoda

(2) Mollusca

(3) Echinodermata

(4) Hemichordata

79) Identify the correct match.

	Column A	**Column B**
(1)	Antedon, Echinus	Exclusively Fresh water forms
(2)	Apis, Laccifer	Economically important insects
(3)	Pheretima, Hirudinaria	External fertilization in water
(4)	Ascaris, Wuchereria	Ectoparasites

(1) 1

(2) 2

(3) 3

(4) 4

80) How many in the given examples of animals are Coelentrates.
Physalia, Obelia, Plannaria, Pennatula, Gorgonia, Pleurobrachia, Meandrina and Nereis

(1) Three

(2) Four

(3) Five

(4) Six

81) Which of the following animal is considered as "Living fossil" :-

(1) *King crab*

(2) Spider

(3) *Periplanata*

(4) Scorpion

82) In which animal pouched mesoderm is present :-

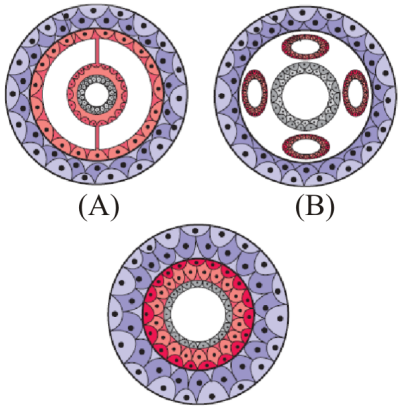
(1) Fasciola

(2) Neries

(3) Ancylostoma

(4) Ctenoplana

83) The figure shows (A, B, C) Three types of conditions in animal. Select the option giving correct identification together with animals.



(1) C-Acoelomate in round worm and earth worm

(2) A-Coelomate in earthworm and cockroach

(3) B-Coelomate in Tapeworms and round worm

(4) A-Pseudocoelomate in round worms

84) **Statement–**I **:-** All Platyhelminthes are endoparasites.
Statement–II **:-** Tapeworm is endoparasite.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statements I is correct

(4) Only statements II is correct

85) In the phylum Mollusca, the body is divided into:

(1) head, thorax and abdomen.

(2) head, muscular foot and abdomen.

(3) head, thorax and visceral hump.

(4) head, muscular foot and visceral hump.

86) Choose the incorrect statement.

(1) Sensory organs like antennae, eyes (compound and simple), statocysts or balancing organs are present in arthropods.

(2) A well developed excretory organs are present in echinoderms.

(3) Jointed appendages is a characteristic feature of phylum Arthropoda.

(4) The organisms of 2nd largest phylum exhibit the same type of symmetry as that of larval forms of echinoderms.

87) Which one of the following option correct for aschelminthes :-

(1) Sexes are not separate (monoecious)
(2) Roundworms have organ level of body organisation and acoelomate animals
(3) Alimentary canal is complete with a well- developed muscular pharynx
(4) Fertilisation is external and development may be direct

88) Match the columns and choose the correct answer from the options given below

COLUMN (A)		**COLUMN (B)**	
A	Triploblastic	i	Mesodermal in origin
B	Notochord	ii	Three germ layers
C	Metamerism	iii	true segmentation
		iv	Chordates

(1) A -ii, B-iii, C-iv
(2) A-i, B-iii, C-iv
(3) A -ii, B-iv & i, C-iii
(4) A-iii, B-ii, C-i

89) The statement which does not stand true for arthropods is

(1) They have an open circulatory system
(2) Their body is divided into head, thorax and abdomen
(3) They are segmented and coelomate animals
(4) Fertilisation is usually external in them

90) Digestion in coelenterates is

(1) Only intracellular
(2) Only extracellular
(3) Both extracellular and intracellular
(4) Not required

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	1	1	1	1	3	4	3	4	2	2	3	4	4	2	3	2	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	4	1	2	1	1	2	3	2	2	1	1	2	3	1	3	4	3	1	3
Q.	41	42	43	44	45															
A.	3	2	2	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	4	2	4	4	3	2	2	2	1	2	3	1	4	1	2	2	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	4	2	2	2	4	4	4	2	4	2	2	4	1	1	3	2	3	1
Q.	86	87	88	89	90															
A.	2	4	3	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	4	3	3	1	2	3	4	4	4	2	2	1	4	2	4	1	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	1	1	2	1	2	4	1	3	4	2	4	2	3	2	2	2	4	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	3	3	2	3	3	4	1	1	2	2	3	3	4	2	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	1	3	2	1	2	4	1	3	2	1	2	3	1	3	4	2	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	2	4	4	2	3	3	4	3										

SOLUTIONS

PHYSICS

1)

Correct Answer is Option (2).

2)

Correct Answer is Option (1).

3)

Binding energy of $_{11}N^{23}$ nucleus
$= 0.21 \times 931.5$ MeV
Binding energy per nucleon

$= \frac{0.21 \times 931.5}{23} = 8.505$ MeV

4) $_{11}Na^{24} \rightarrow {}_{11}Na^{23} + {}_{0}^{1}n$
$\Delta = 23 \times 7.68 - 24 \times 7048 = -2.88$ MeV

5) $\frac{hc}{\lambda} = k + \phi$ .....1 $\frac{2hc}{\lambda} = 3k + \phi$ 2
Solve equation (1) and (2)

6) $E\lambda_1 = \frac{he}{\lambda} = \frac{1240}{550}eV = 2.25eV$
$E\lambda_2 = \frac{1240}{450}eV = 2.8eV$
and $E\lambda_3 = \frac{1240}{250}eV = 3.5eV$
For metal r, λ_3 is able generate photoelectron
For metal q, λ_2, λ_3 are able to generate electron
For metal p, all wavelength are able to generate electrons.
Hence photoelectric current will be maximum for P and minimum for r.

7) $Power = W = \frac{n_{ph}}{t} \times E$
$W = \frac{10^{20}}{16} \times 7 \times 1.6 \times 10^{-19}$Watt

W = 7 Watt

8) $\lambda = \frac{hc}{\Delta E} = \frac{hc}{2E - E}$

$\lambda = \frac{hc}{E}$ --------(1)

for next

$\Delta E = \frac{4E}{3} - E = \frac{E}{3}$

$\lambda' = \frac{hc}{\frac{E}{3}} = \frac{3hc}{E}$

$=3\lambda$

9) ${}_{90}Th^{220} \rightarrow {}_{2}He^{4} + {}_{88}Ra^{216}$

$KE = \frac{P^2}{2m}$

$\frac{KE_{Ra}}{KE_{\alpha}} = \frac{4}{A-4}$

$KE_{Ra} = \frac{4E_{\alpha}}{220-4} = \frac{E_{\alpha}}{54}$

10) $r = r \cdot \frac{n^2}{2}$

for Li^{2+}

$r_5 = r \cdot \frac{25}{3}$

for He^{+}

$r_5 = r \cdot \frac{25}{2}$

$\therefore \frac{r_{Li^{2+}}}{r_{He^{+}}} = \frac{2}{3}$

11)

Correct Answer is Option (2).

12)

Correct Answer is Option (2).

13)

Correct Answer is Option (3).

14)

Correct Answer is Option (4).

15) $F = \frac{(1 + r)IA}{C}$

16) From $eV_0 = h\nu - \phi$ and $K_{max} = h\nu - \phi$. If ν increases keeping ϕ constant, then V_0 and K_{max} both increase.
If ϕ decreases keeping ν constant, then V_0 and K_{max} increase.
If I increases more photoelectrons would be liberated, hence saturation photocurrent increases.
If separation between cathode and anode is increased, then there is no effect on ν_0, K_{max} or current.

17)

Radiation force on boat = $\frac{2IA}{C} = \frac{2PA}{4\pi r^2 C}$

Acceleration = $\frac{PA}{2\pi r^2 Cm}$

18) Total number of photons emitted per second from the light source

$$N = \frac{\text{Power of the source}}{\text{Energy of each photon}} = \frac{1.0 \text{ J/s}}{\frac{hc}{\lambda}\text{J}}$$

$$N = \frac{\lambda \times 1.0}{hc} = \frac{4.8 \times 10^{-7}}{6.6 \times 10^{-34} \times 3 \times 10^{8}} = 2.424 \times 10^{18}$$

These photons spread out in all directions. The number of photons striking on 1 m^2 surface per second is

$$n = \frac{N}{4\pi r^2} = \frac{2.424 \times 10^{18}}{4 \times 3.14 \times (2.0)^2} = 4.825 \times 10^{16}$$

19) ${}_{90}Th^{220} \rightarrow 2He^4 + 88Ra^{216}$

$m_\alpha v_\alpha = m_R v_R$

$$\frac{v_R}{v_\alpha} = \frac{1}{54}$$

20)

Correct Answer is Option (2).

21)

Question Explanation : Which of the given relations can be derived using dimensional analysis.

Concept : This question is based on limitations of dimensional analysis.

Solution :
1. $v^2 u^2 + 2as \rightarrow$ involves addition; dimensional analysis can't confirm additive form.
2. $y = A \cos(\omega t - kx) \rightarrow$ involves trigonometric function,so relations involving trigonometric function can't be derived using dimensional analysis

3. $A = A_0e^{-\lambda t}$ → involves exponential function,so relations involving exponential function can't be derived using dimensional analysis
Hence, **none** of these can be derived purely by dimensional analysis.

Final answer : Option (4)

22)

Question Explanation : Given that [A] ≠ [B], We have to check which operation given in the options is valid.

Concept : This question is based on law of Homogeneity of Dimensions.

Solution :
The argument of log and exponential functions should be dimensionless
Let's check each option one by one.

Options 1 : log (A/B) : $\left[\frac{A}{B}\right]$ = should be Dimension less ⇒ but [A] ≠ [B] (doesn't match with given condition)

Option 2 : $(e)^{A/B} \Rightarrow \left[\frac{A}{B}\right]$ = should be Dimension less ⇒ but [A] ≠ [B] (doesn't match with given condition)

Option 3 : A + B ⇒ □ [A] ≠ [B] (doesn't match with given condition since two quantities with different dimensions can't be added together)

Option 4 : A × B : ⇒ This operation is valid even if [A] may or may not be equal to [B] as it gives a new dimension ⇒ [AB] (matches with given condition).

Final answer : Option (4)

23)

Correct Answer is Option (1).

24) **Question Explanation :** Asking about x in which diameter reading of wire is asking in screw gouge least cant and M.S.R are given.
Concept : Screw gauge
Solution : Diameter = $\text{M.S.R.} + (\text{f} \times \text{L.C.})$

Diameter = $1\text{ mm} + 42 \times \frac{1\text{mm}}{100}$

Diameter = $\frac{142}{100}\text{mm}$

Diameter = $\frac{71}{50}\text{mm}$

⇒ So ; x = 71

Final Answer : Option (2)

25)

Explain Question: Find ratio of admissions of given physical question.

Concept: Dimension analysis

Solution:

$$F = \frac{q^2}{4\pi \in_0 r^2}$$

$$[\in_0] = \left[\frac{q^2}{Fr^2}\right] = \left[\frac{A^2T^2}{MLT^{-2}.L^2}\right] = \left[M^{-1}L^{-3}T^4A^2\right]$$

$$c = \frac{1}{\sqrt{\mu_0 \in_0}} \Rightarrow \mu_0 = \frac{1}{\in_0 c^2}$$

$$\frac{\in_0}{\mu_0} = \frac{\in_0}{1/(\in_0 c^2)} = \in_0^2 c^2$$

$$\left[\frac{\in_0}{\mu_0}\right] = \left[M^{-1}L^{-3}T^4A^2\right]^2\left[L^1T^{-1}\right]^2 = \left[M^{-2}L^{-4}T^6A^4\right]$$

Final answer: Option (1)

26)

Explain the Question : The position of a particle is given by $x(t) = \frac{v_0}{\alpha}(1 - e^{-\alpha t})$ where v_0 is a constant and $\alpha > 0$. find the dimensions of v_0 and α.

Concept : This question is based on dimensional Homogeneity (Principle of Homogeneity).

Solution : Exponents must be dimensionless

$[\alpha t] = [M^0L^0T^0]$

$[\alpha] = [M^0L^0T^{-1}] = [T^{-1}]$

Now, $[x] = \frac{[v_0]}{[\alpha]}$

$[v_0] = [x]\,[\alpha]$

$[v_0] = [L]\,[T^{-1}] = [LT^{-1}]$

Final answer : Option (1)

27) **Question Explanation:** In this question, we have to find the dimensional formula of energy in terms of force, acceleration and time.

Concept: Dimensional analysis (deducing relation between known quantities)

Solution: Let assume

$E \propto F^a A^b T^c$

$E = KF^aA^bT^c$ (K = Dimensionless constant)

Putting the dimensional formula of given physical quantities.

$$\left[ML^2T^{-2}\right] = \left[MLT^{-2}\right]^a\left[LT^{-2}\right]^b[T]^c$$

$\left[ML^2T^{-2}\right] = \left[M^aL^{a+b}T^{-2a-2b+c}\right]$

By comparison, we get

a = 1, a + b = 2 ⇒ b = 1

and –2a – 2b + c = –2 ⇒ c = 2

 E = $KFAT^2$

Or E = FAT^2 (assuming K = 1)

Final Answer: Option (2)

28)

Question Explanation :

$Q = \frac{A^3B^3}{C\sqrt{D}}$

$\left(\frac{\Delta A}{A} \times 100\%\right)$ = 2%

$\left(\frac{\Delta B}{B} \times 100\%\right)$ =1%

$\left(\frac{\Delta C}{C} \times 100\%\right)$ = 3%

$\left(\frac{\Delta D}{D} \times 100\%\right)$ = 4%

$\left(\frac{\Delta Q}{Q} \times 100\%\right)$ = ?

Concept : Error in multiplication, division & exponent.

Solution :

$Q = A^3B^3C^{-1}D^{-\frac{1}{2}}$

$\left(\frac{\Delta Q}{Q} \times 100\right) = 3\left(\frac{\Delta A}{A} \times 100\right) + 3\left(\frac{\Delta B}{B} \times 100\right) + \left(\frac{\Delta C}{C} \times 100\right) + \frac{1}{2}\left(\frac{\Delta D}{D} \times 100\right)$

$\left(\frac{\Delta Q}{Q} \times 100\right)$ = + 3 (2%) + 3 (1%) + 1 (3%) + $\frac{1}{2}$ (4%)

$\left(\frac{\Delta Q}{Q} \times 100\right)$ = 6% + 3% + 3% + 2% = 14%

Final Answer : option (3)

29) Average value $= \frac{2.63 + 2.56 + 2.42 + 2.71 + 2.80}{5}$

= 2.62 sec.

Now $|\Delta T_1| = 2.63 - 2.62 = 0.01$

$|\Delta T_2| = 2.62 - 2.56 = 0.06$

$|\Delta T_3| = 2.62 - 2.42 = 0.20$

$|\Delta T_4| = 2.71 - 2.62 = 0.09$

ΔT_5 = 2.80 – 2.62 = 0.18

Mean absolute error

$$\Delta T = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4| + |\Delta T_5|}{5}$$

$$= \frac{0.54}{5} = 0.108 = 0.11 \text{sec}$$

30) 1MSD = 1 mm; Given : 20VSD = 16 MSD

L.C. : = 1MSD – 1VSD $1\text{VSD} = \frac{16}{20}$ MSD

= 1 mm –.8 mm = .2 mm

31)

Explain Question :

A science student Performs on experiment twice –

first Time :–

No. of observations$\Rightarrow N_1 = 100$

Random error $\Rightarrow \Delta E_1 = x$

Second Time :–

No of Observations $\Rightarrow N_2 = 600$

Random error $\Rightarrow \Delta E_2 = ?$

Concept :

Random errors

Formula:

Random error $\propto \frac{1}{\text{No. of Observations}}$

Solution :

$$\Delta E \propto \frac{1}{N}$$

$$\Rightarrow \frac{\Delta E_1}{\Delta E_2} = \frac{N_1}{N_2}$$

$$\Rightarrow \frac{\Delta E_2}{x} = \frac{100}{600}$$

$$\Rightarrow \Delta E_2 = \frac{1}{6} \times x$$

$\Rightarrow \Delta E_2 = \frac{1}{6}$ times of initial value

Final Answer : option (1)

32)

Correct Answer is Option (1).

33) **Explain question:**

Calculate volume of block and then round the answer according to significant rules.

Concept:
Significant figures

Solution:
Volume = $V = \ell \times b \times t$
$= 12 \times 6 \times 2.45 = 176.4\ cm^3$
$V = 1.764 \times 10^2\ cm^3$
since, the minimum number of significantfigure is one in breadth, hence volume will also contain only one significant figure. Hence, $V = 2 \times 10^2\ cm^3$.

Final answer: option (2)

34)

Correct Answer is Option (3).

35)

For a given source,
Intensity α number of photons per sec.
α number of photo e^- per sec.
α current

36)

Explanation Question:
The question asks about the stability of nucleons (protons and neutrons) when they exist outside of an atomic nucleus.

Concept:
Radioactive decay of free nucleons.

Solution:
Inside a nucleus, neutrons are generally stable due to nuclear forces.
However, outside the nucleus, a free neutron is unstable.
A free neutron undergoes beta decay to transform into a proton, an electron, and an antineutrino.
$n \rightarrow p + e^- + v_e$,
The mean life of a free neutron is approximately 1000 seconds.
In contrast, a free proton is considered stable.
Its predicted half - life is extremely long, exceeding the current age of the universe.
Therefore, outside the nucleus, the neutron is unstable.

Final answer : Option (3)

37)

Explain Question : Lenard observed that photoelectric emission occurs only when the

frequency of incident light is greater than a certain minimum value, called the threshold frequency. We have to identify what this minimum frequency depends on.

Concept : This question is based on The threshold frequency is related to the work function of the metal. The work function is a property of the emitting material and does not depend on external experimental conditions.

Solution :

Threshold frequency $\upsilon_0 = \frac{\phi}{h}$
The work function ϕ depends only on the material of the emitter plate
It is independent of potential difference, plate separation, or size of the plate

Final answer : Option (4)

38)

Explain Question : Thermonuclear fusion requires extremely high temperature so that positively charged nuclei can come close enough to fuse. We have to identify the incorrect statement related to thermonuclear fusion.
Concept : This question is based on At very high temperature, nuclei gain large kinetic energy. This energy helps them overcome Coulomb repulsion and allows fusion to occur. Such reactions take place in stellar interiors at temperatures of the order of 10^8 K.
Solution :
Consider each statement:
1. Energy is provided by raising the temperature.
This is correct because increasing temperature increases the kinetic energy of nuclei.
2. Energy must be provided to overcome Coulomb repulsion between positively charged nuclei.
This is correct because like charges repel each other.
3. Is responsible for the formation of red giant star.
This is incorrect. Red giant formation mainly occurs due to gravitational effects and changes in nuclear fuel, not directly because of thermonuclear fusion.
4. Required temperature is of the order of 10^8 K.
This is correct for thermonuclear fusion.
Final answer : 3

39) **Explanation Question:**
Calculate the duration for which a 100W lamp can be powered by the fusion of 2kg of deuterium.

Concept: Time duration $(t) = \frac{\text{Total Energy}}{\text{Power}}$

Solution:
Mass of deuterium = 2kg = 2000g
Number of moles = $\frac{2000}{2}$ = 1000 moles
Number of nuclei N = $1000 \times 6.023 \times 10^{23}$

$N = 6.023 \times 10^{26}$
Two deuterium nuclei are required per reaction.
Total reactions = $N / 2 = 3.2115 \times 10^{26}$
Total energy $E = 3.0115 \times 10^{26} \times 3.27 \times 106^{6} \times 1.6 \times 10^{-19}$
$E = 1.575 \times 10^{14}$ J
Time $t = E / P = 1.575 \times 10^{14} / 100$
$t = 1.575 \times 10^{12}$ s
t in years $= 1.575 \times 10^{12} / (3.1536 \times 10^{7})$
$t = 4.9 \times 10^{4}$ year.

Final answer : Option (1)

40)

Explain Question : The radius of a nucleus depends on its mass number. Using the given data and the relation between radius and mass number, we have to find the value of x.
Concept : This question is based on $R = R_0A^{1/3}$
So, $R^3 \propto A$
Solution :
Given :
$R_1 = 4.8$ fm, $A_1 = 64$
$R_2 = 4$ fm, $A_2 = A$
Using $R^3 \propto A$

$$\left(\frac{4.8}{4}\right)^3 = \frac{64}{A}$$

$$(1.2)^3 = \frac{64}{A}$$

$$1.44 \times 1.2 = \frac{64}{A}$$

$$A = \frac{64}{1.44 \times 1.2}$$

$$A = \frac{1000}{x}$$

$$x = \frac{144 \times 12}{64} = 27$$

Final answer : 3

41) $P = nh\nu$

$$n = \frac{P}{h\nu} = \frac{2 \times 10^{-3}}{6.63 \times 10^{-34} \times 6 \times 10^{14}}$$

$= 5 \times 10^{15}$

42) **Question Explanation**
In this question, we are asked to find the unknown nucleus X using conservation of mass number and atomic number.

Concept: Nuclear reaction (conservation of mass number and atomic number.)

Solution:

In any nuclear reaction, the total sum of the mass numbers and the total sum of the atomic numbers must remain constant on both sides of the equation.

1. Conservation of Mass Number $(A): \sum A_{reactants} = \sum A_{products}$

2. Conservation of Atomic Number (Z): $\sum Z_{reactants} = \sum Z_{products}$

Given reaction: ${}_{7}^{14}N + {}_{2}^{4}He \rightarrow X + {}_{1}^{1}H$

Mass number conservation: $14 + 4 = A_X + 1 \Rightarrow A_X = 17$

Atomic number conservation: $7 + 2 = Z_X + 1 \Rightarrow Z_X = 8$

Element with Z = 8 is Oxygen.

So: $X = {}_{8}^{17}O$

Final answer : Option (2)

43)

Explain question:

The experiment describes a setup where a collector plate is positioned vertically above an emitter plate.

After reaching a saturation photo current, a vertically downward electric field is switched on.

We need to determine how this field affects the electrons.

Concept: Photoelectric Effect: Motion of charged particles in an electric field (F = qE).

Solution:

1. Force Direction: The electric field is downward, so the force on the electrons is upward.

2. Kinetic Energy: The electrons are already moving upward toward the collector. The upward force accelerates them,

causing their kinetic energy to increase by the time they hit the plate.

3. Other Parameters: Stopping potential and threshold wavelength are properties of the light and metal surface, not the external field.

Final answer: Option (2)

44) **Explain Question:**

The question examines the correctness of an assertion about the energy-momentum relation of a photo and whether the given reason correctly explains it.

Concept:

Photon shows particle nature and hence carries momentum.

Solution:

For a photon:

$$E = pc \Rightarrow p = \frac{E}{c}$$

So, the Assertion is correct.

A photon behaves like a particle and therefore possesses momentum, which explains why it follows the relation $p = \frac{E}{c}$. Hence, the Reason is also correct and properly explains the assertion.

Final answer: Option (1)

45) $\frac{mv^2}{a_0} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{a_0^2} \Rightarrow v = \frac{e}{\sqrt{4\pi\varepsilon_0 a_0 m}}$

CHEMISTRY

46)

Explaining : -
Concept : This question is based on ionic size
Solution :
For isoelectronic species
Ionic size $\propto \frac{\text{Negative charge}}{\text{Positive charge}}$, P^{-3}, N^{-3} on going top to bottom , size increases.
Final Answer : (1)

47)

Explaining : -
Concept : This question is based on atomic radius
Solution :
For the cationic species of same element.
Atomic radius $\propto \frac{1}{Z_{eff}} \propto \frac{1}{\text{Positive charge}}$

+2	+3	+4	+6
MnO	Mn_2O_3	MnO_2	MnO_3

In MnO_3, Mn^{+3} is having least ionic radius.
Final Answer : (3)

48)

Explaining : -
Concept : This question is based on ionisation potential
Solution : Number of IP values before sudden rise in IP sudden rise in IP is 19 → 120 i.e. 1st to 2nd I.P., Hence there is one electron in valence shell, so correct element is Na.
Final Answer : (4)

49)

Explaining : -
Concept : This question is based on
Solution : Ionisation potential, Ne[$3s^2 3p^3$] is Phosphours having highest ionisation energy in given options.
Final Answer : (2)

50)

Explaining :
Concept : This question is based on ionisation potential
Solution : Be is having higher I.P. than the I.P. of given element
$IP \propto \frac{1}{size}$
Be > B > Li > Ag > Al > K.
Final Answer : (4)

51)

Explaining : -
Concept : This question is based on electron affinity.
Solution :
Generally
Electron affinity $\propto \frac{Z_{eff}}{size}$, But correct EA order N < P < C < Si
EA : P > N
$\because \Delta H_{eq}$ for P is negative
ΔH_{eq} for N is positive
Final Answer : (4)

52) **Explaining :** -
Concept : This question is based on electronaffinity and ionisation potential.
Solution :
$X^- \rightarrow F^-$ and $Y^+ \Rightarrow Na^+$?
I.P. of F > |ΔHeg $Na^+(g)$|
Final Answer : (3)

53)

Correct Answer is Option (2).

54) **Explain Question:** Identify Amphoteric oxide
Concept: Nature of oxides
Solution:
CO_2, SiO_2 → acidic
CaO → basic
$A\square_2O_3$ → Amphoteric oxide
Final Answer (2)

55)

Correct Answer is Option (2).

56) All the species have 10 electrons each and hence are isoelectronic ions. Among

isoelectronic ions, ionic size increase as the +ve charge decreases or the –ve charge increases, i.e., option (1) is correct.

57)

Correct Answer is Option (2).

58) XI NCERT Page No. 95

59)

Correct Answer is Option (1).

60)

Correct Answer is Option (4).

61) Metallic character order :
Na > Mg > Be > Si > P

62) **Asking About:**
• Determining the element based on its successive ionization energies.
Concept: (Identification of element based on IE)
• lonization energies increase as successive electrons are removed.
• Large jumps in ionization energies indicate the removal of an electron from a core shell, which is much more difficult.
• The large jump in ionization energies indicate the number of valence electrons.
Solution/Explanation:
• Analyzing the given ionization energies:
7.1 eV (IP_1)
14.3 eV (IP_2)
34.5 eV (IP_3)
46.8 eV (IP_4)
162.2 eV (IP_5)
• There is a significant jump between IP_4 (46.8 eV) and IP_5 (162.2 eV). This indicates that the element has 4 valence electrons.
• Silicon (Si) has 4 valence electrons.
Answer: Option 2

63) **Question Explanation :**
Question is asking about calculating the total enthalpy required for the ionization of magnesium to Mg^{2+}

Concept:
This question is based on lonization enthalpy (IP) is the energy required to remove an electron from a gaseous atom or ion.

The overall enthalpy change for the reaction is the sum of the individual ionization enthalpies.

Formula:
ΔH (Enthalpy change) = $IP_1 + IP_2$

Solution :
The reaction $Mg \rightarrow Mg^{2+} + 2e^-$ involves two steps:
$Mg \rightarrow Mg^+ + +e^-$ (IP_1 = 178 K. cal mol^{-1})
$Mg^+ \rightarrow Mg^{2+} + e^-$ (IP_2 = 348 K. cal mol^{-1})
To find change, we add IP_1 and IP_2
$\Delta H = 178 + 348 = 526$ K. cal mol^{-1}
Since ionization is an endothermic process, the enthalpy change is positive.

Final Answer :
Option (2)

64)

Asking About : Comparing the screening effect of d-electrons to p-electrons and f-electrons.

Concept :
The screening effect is the ability of inner-shell electrons to reduce the effective nuclear charge experienced by outer-shell electrons
The effectiveness of screening depends on the shape of the orbital
s > p > d > f (in general screening order)

Solution :
This is due to the shapes of the orbitals and their penetration towards the nucleus
therefore d-electrons have less screening effect than p-electrons.
(d-electrons have complicated and diffused shape so penetration is less)

Answer option (4) correct

65)

Correct Answer is Option (4).

66)

Correct Answer is Option (4).

67)

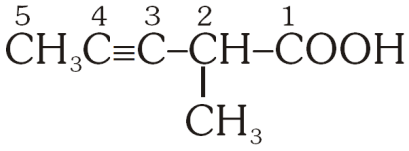
Correct Answer is Option (3).

68)

Correct Answer is Option (4).

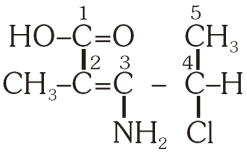
69)

Correct Answer is Option (2).



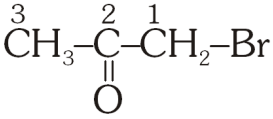
70)

4-Methyl pentanal

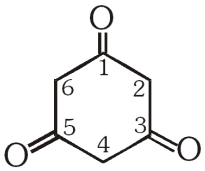


71)

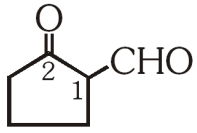
3-Amino-4-chloro-2-methylpent-2-enoic acid



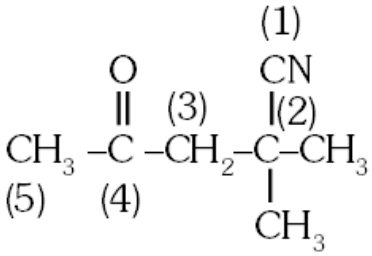
72) (1)



(2)

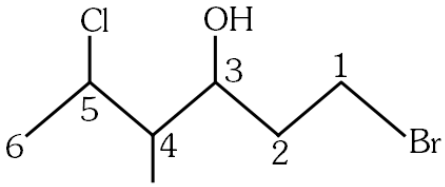


(3)



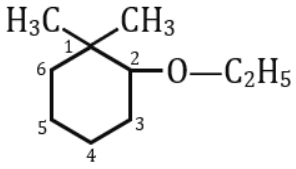
73)

2,2-Dimethyl-4-oxopentanenitrile



74)

1-Bromo-5-chloro-4-methylhexan-3-ol

75)  2-Ethoxy-1, 1-dimethyl cyclohexane

76)

Correct Answer is Option (4).

77)

Correct Answer is Option (2).

78)

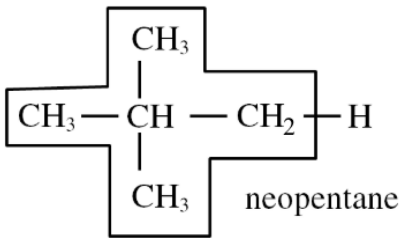
Correct Answer is Option (2).

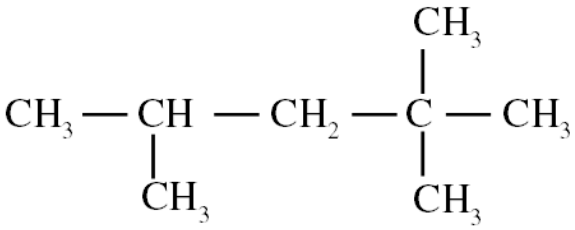
79)

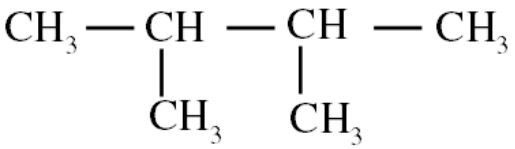
Question Explanation: Asked which structures do not use the prefix "Neo".

Concept: Alkane nomenclature, Neo prefix rule, Branched chain recognition.

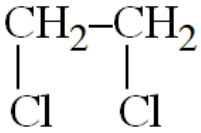
Solution:

(1)  neopentane

(2)  Iso-Octane (Exception)

(3)  No Quaternary Carbon Present → Not Neo.

Final Answer: Option (4)

80) 
ethylene chloride (no π bonds)

81)

Correct Answer is Option (1).

82)

Correct Answer is Option (3).

83)

Correct Answer is Option (2).

84)

Correct Answer is Option (3).

85)

Correct Answer is Option (1).

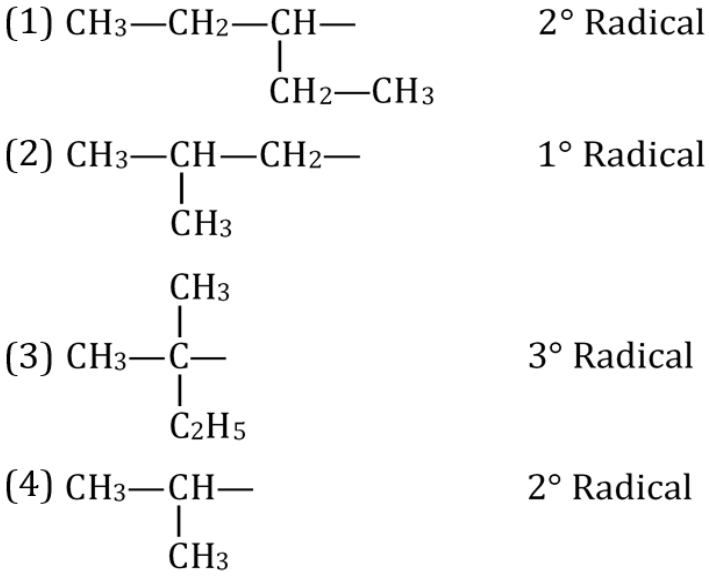
86)

Correct Answer is Option (2).

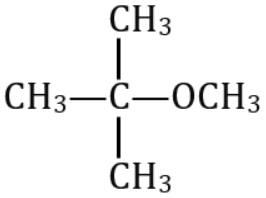
87)

Correct Answer is Option (4).

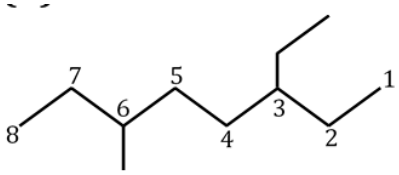
88)



89) Tertiary ether has alkoxy group (RO—) attached to tertiary carbon.



90)



3-ethyl-6-methyloctane

BIOLOGY

91)

Correct Answer is Option (2).

92)

Correct Answer is Option (1).

93)

Correct Answer is Option (4).

94)

Correct Answer is Option (3).

95) NCERT-XI, Pg#07

96)

Correct Answer is Option (1).

97)

Correct Answer is Option (2).

98)

Correct Answer is Option (3).

99)

Correct Answer is Option (4).

100)

Correct Answer is Option (4).

101) Correct Answer is Option (4).

102)

Correct Answer is Option (2).

103) NCERT Pg. # 6

104)

Question Explanation:
Correct rules of nomenclature.

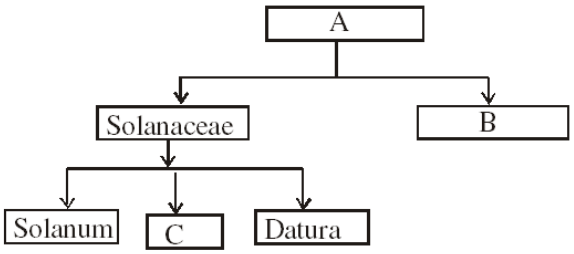
Concept:
Binomial Nomenclature Rules

Solution:
Both the genus and specific epithet are italicized when printed, or underlined when handwritten.

Final Answer: Option 1-Both genus and specific epithet are printed in italics

105)

Explain Question: The question is asking to identify the correct taxonomic ranks for the labels in the diagram.



Concept: Taxonomic categories

Solution: Based on the diagram and classification: A → Order → Polymoniales B → Family → Convolvulaceae C → Genus → Petunia

Explanation:Plant families like Solanaceae and Convolvulaceae are grouped under the order Polymoniales because of shared floral characteristics.Within the family Solanaceae the genus Petunia is an example of a member genus.
Final answer: option 4

106)

Explain Question: The question asks for the taxonomic category defined as a collection of species showing close resemblance

Concept: Genus

Solution: A collection of species with close resemblance in floral morphology is called a genus.

Final answer: 2

107)

Correct Answer is Option (4).

108)

Correct Answer is Option (1).

109)

Correct Answer is Option (2).

110)

Correct Answer is Option (1).

111)

Correct Answer is Option (3).

112)

Correct Answer is Option (1).

113)

Correct Answer is Option (1).

114)

Correct Answer is Option (2).

115)

Correct Answer is Option (1).

116) NCERT-XI, Pg. # 05

117)

Correct Answer is Option (4).

118) NCERT-XI Pg. # 6

119)

Correct Answer is Option (3).

120)

Correct Answer is Option (4).

121)

In Carl Woese's classification system, the kingdom that includes fungi is Eukarya.

The correct answer is: OPTION (2) Eukarya

122)

Correct Answer is Option (4).

123)

Correct Answer is Option (2).

124)

Correct Answer is Option (3).

125) NCERT Pg. # 16

126)

Correct Answer is Option (2).

127)

Correct Answer is Option (2).

128)

Correct Answer is Option (4).

129)

Correct Answer is Option (3).

130)

Correct Answer is Option (4).

131)

Both statements are incorrect.

132)

Correct Answer is Option (3).

133)

Correct Answer is Option (4).

134)

Correct Answer is Option (3).

135)

Correct Answer is Option (3).

136) Echinoderms are exclusively marine, mostly porifera are assymmetrical., all ctenophore are marine, some annelids are perositic.

137) NCERT

138) NCERT Exercise

139)

Correct Answer is Option (4).

140)

Correct Answer is Option (1).

141)

Correct Answer is Option (1).

142)

Correct Answer is Option (2).

143) **Question Explanation:**
This question is asking to correctly match the given animals in Column-I with their corresponding common examples or habitat forms in Column-II.
Concept :
Animal Kingdom: Topic – classification and examples of Porifera and Cnidaria.

Explanation:

A. Spongilla → fresh water sponge → (A – iii)
B. Gorgonia → sea fan → (B – i)
C. Euspongia → bath sponge → (C – v)
D. Meandrina → brain coral → (D – ii)
E. Pennatula → sea pen → (E – iv)

Thus, the correct matching is A-iii, B-i, C-v, D-ii, E-iv.

Final Answer: Option 2

144) NCERT Pg # 54

145) **NCERT Reference:**
Zoology-XI Page No.-51

146) **NCERT Reference:**
Zoology-XI Page No.-51, 53, 54

147) **NCERT Reference:**
Zoology-XI Page No.-60

148) **NCERT Reference:**
Zoology-XI Page No.-51

149) **NCERT Reference:**
Zoology-XI Page No.- 52, 53, 54

150) **NCERT Reference:**
Zoology-XI Page No.-49, 50, 51, 52, 53, 54

151) NCERT Pg.#53, Para-4.2.7

152)

Correct Answer is Option (3).

153)

Correct Answer is Option (1).

154)

Correct Answer is Option (3).

155) NCERT Pg. # 54

156) **Explain Question :** This question is asking about: Feeding organ of Mollusca.
Concept : Animal Kingdom → Phylum Mollusca.
Solution : Radula is a rasping, file-like organ present in Mollusca, used for scraping and feeding.
So, both assertion and reason are true, and the reason correctly explains assertion.

Final answer : Option (1)

157)

Correct Answer is Option (2).

158)

Correct Answer is Option (4).

159)

Correct Answer is Option (1).

160)

Correct Answer is Option (3).

161) **Question Explanation:** Relationship between tissue and organs.
Concept: Levels of Organization
Solution: The **Assertion** is true; tissues are indeed grouped to form organs with specialized functions. However, the **Reason** is also true, as the phylum **Platyhelminthes** is the first group to exhibit the **organ level of organization**. But the Reason does not explain the Assertion; the Reason is a separate fact about Platyhelminthes, not an explanation of why tissues form organs.
Final Answer: Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

162) NCERT Pg. # 48

163) **Question Explanation**: Type and location of fertilization in sponges.
Concept: Porifera Reproduction.
Solution: Sponges are mostly hermaphrodites, but fertilization is **internal** (inside the body) and involves sperm from another individual (**cross-fertilization**).
Final Answer: 2

164) **Question Explanation:** Germ layer terminology
Concept: Number of germinal layers
Solution: Diploblastic and triploblastic describe the number of germinal layers (two or three) present during embryonic development in animals.
Final Answer: The number of germinal layers during embryonic development

165) **Question Explanation**: The question asks to determine the correctness of the two statements regarding types of circulatory systems — open and closed circulation.

Concept: Circulatory pathways

Solution:

A. Statement I: Correct — describes open circulation accurately.

B. Statement II: Correct — describes closed circulation accurately.

Final Answer: Option – 1 → Both statement-I and statement-II are correct.

168)

Correct Answer is Option (2).

169)

Correct Answer is Option (2).

171)

Question Explanation: This question asks which of the following animals is considered a "living fossil."

Concept: Arthropoda.

Solution: The correct answer is Option 1) Limulus

A. Limulus (horseshoe crab) is often called a "living fossil" because it has remained largely unchanged for over 450 million years.

Final Answer: Option (1)

172)

NCERT-XI, Pg # 48

173)

Correct Answer is Option (2).

174)

Explain Question : This question is asking about whether Platyhelminthes are all endoparasites and whether tapeworm is an endoparasite.

Concept : Animal Kingdom – Phylum Platyhelminthes.

Explanation : Not all Platyhelminthes are endoparasites; some (like Planaria) are free-living. Tapeworm (Taenia) is an endoparasite. Thus, only statement II is correct.

Final answer : Option 4

175) NEW NCERT 11th Page No. 44

176) NEW NCERT 11th Page No. 45

177)

Correct Answer is Option (3).